

Overview of materials for Nylon 610

Categories: [Polymer](#); [Thermoplastic](#); [Nylon \(Polyamide PA\)](#); [Nylon 610 \(PA610\)](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Nylon 610". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Click here to view all available suppliers for this material.](#)

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Physical Properties	Metric	English	Comments
Density	0.930 - 3.80 g/cc	0.0336 - 0.137 lb/in ³	Average value: 1.26 g/cc Grade Count:265
Filler Content	15.0 - 50.0 %	15.0 - 50.0 %	Average value: 28.9 % Grade Count:41
Water Absorption	0.100 - 4.80 %	0.100 - 4.80 %	Average value: 1.45 % Grade Count:127
Moisture Absorption at Equilibrium	0.100 - 1.80 %	0.100 - 1.80 %	Average value: 0.703 % Grade Count:86
 Water Absorption at Saturation	0.900 - 1.70 % @Temperature 70.0 - 70.0 °C	0.900 - 1.70 % @Temperature 158 - 158 °F	Average value: 1.27 % Grade Count:18
 Water Absorption at Saturation	0.350 - 6.00 %	0.350 - 6.00 %	Average value: 2.68 % Grade Count:53
 Water Absorption at Saturation	0.400 - 1.20 % @Temperature 45.0 - 400 °C	0.400 - 1.20 % @Temperature 113 - 752 °F	Average value: 0.814 % Grade Count:2
 Water Absorption at Saturation	0.400 - 1.20 % @Treatment Temp. 70.0 - 70.0 °C	0.400 - 1.20 % @Treatment Temp. 158 - 158 °F	Average value: 0.814 % Grade Count:2
Additive Loading	10.0 - 40.0 %	10.0 - 40.0 %	Average value: 24.5 % Grade Count:4
Viscosity	360000 - 360000 cP @Temperature 250 - 250 °C	360000 - 360000 cP @Temperature 482 - 482 °F	Average value: 301000 cP Grade Count:1
 Viscosity	0.200 - 1.30e+6 cP @Temperature 230 - 300 °C	0.200 - 1.30e+6 cP @Temperature 446 - 572 °F	Average value: 301000 cP Grade Count:8
 Viscosity	0.200 - 1.30e+6 cP @Shear Rate 10.0 - 8000 1/s	0.200 - 1.30e+6 cP @Shear Rate 10.0 - 8000 1/s	Average value: 301000 cP Grade Count:8
Viscosity Test	130 - 150 cm ³ /g	1.30 - 1.50 dl/g	Average value: 148 cm ³ /g Grade Count:11
Viscosity Number	145 - 220 cm ³ /g	1.45 - 2.20 dl/g	Average value: 171 cm ³ /g Grade Count:7
Linear Mold Shrinkage	0.000600 - 0.0510 cm/cm	0.000600 - 0.0510 in/in	Average value: 0.00769 cm/cm Grade Count:193
Linear Mold Shrinkage, Transverse	0.00100 - 0.0790 cm/cm	0.00100 - 0.0790 in/in	Average value: 0.0131 cm/cm Grade Count:129
Melt Flow	4.00 - 130 g/10 min	4.00 - 130 g/10 min	Average value: 39.8 g/10 min Grade Count:17
Spiral Flow	35.0 - 700 cm	13.8 - 276 in	Average value: 110 cm Grade Count:10

Mechanical Properties	Metric	English	Comments
Hardness, Rockwell R	112 - 122	112 - 122	Average value: 118 Grade Count:15
Hardness, Shore D	63.0 - 84.0	63.0 - 84.0	Average value: 73.3 Grade Count:4
Ball Indentation Hardness	25.0 - 245 MPa	3630 - 35500 psi	Average value: 112 MPa Grade Count:51
Tensile Strength, Ultimate	20.0 - 1330 MPa	2900 - 193000 psi	Average value: 127 MPa Grade Count:217
 Tensile Strength, Ultimate	25.0 - 155 MPa @Temperature -40.0 - 150 °C	3630 - 22500 psi @Temperature -40.0 - 302 °F	Average value: 97.0 MPa Grade Count:3
Tensile Strength, Yield	24.0 - 193 MPa	3480 - 28000 psi	Average value: 82.8 MPa Grade Count:66
Elongation at Break	0.500 - 300 %	0.500 - 300 %	Average value: 28.1 % Grade Count:231
 Elongation at Break	3.30 - 6.10 % @Temperature 60.0 - 150 °C	3.30 - 6.10 % @Temperature 140 - 302 °F	Average value: 36.4 % Grade Count:3
Elongation at Yield	1.60 - 100 %	1.60 - 100 %	Average value: 10.8 % Grade Count:66
Modulus of Elasticity	0.350 - 27.6 GPa	50.8 - 4000 ksi	Average value: 6.83 GPa Grade Count:224
 Modulus of Elasticity	0.200 - 9.80 GPa @Temperature 60.0 - 150 °C	29.0 - 1420 ksi @Temperature 140 - 302 °F	Average value: 4.05 GPa Grade Count:3
Tenacity	0.370 - 0.509 N/tex	4.19 - 5.77 g/denier	Average value: 0.447 N/tex Grade Count:3
Flexural Yield Strength	15.0 - 393 MPa	2180 - 57000 psi	Average value: 196 MPa Grade Count:131
Flexural Modulus	0.120 - 22.1 GPa	17.4 - 3200 ksi	Average value: 7.10 GPa Grade Count:153
 Flexural Modulus	0.500 - 2.30 GPa @Temperature -40.0 - 80.0 °C	72.5 - 334 ksi @Temperature -40.0 - 176 °F	Average value: 1.40 GPa Grade Count:1
Flexural Strain at Break	3.60 - 5.50 %	3.60 - 5.50 %	Average value: 5.09 % Grade Count:6
Compressive Yield Strength	65.0 - 179 MPa	9430 - 26000 psi	Average value: 136 MPa Grade Count:7
Poissons Ratio	0.340 - 0.460	0.340 - 0.460	Average value: 0.403 Grade Count:13
Shear Strength	45.0 - 79.0 MPa	6530 - 11500 psi	Average value: 60.3 MPa Grade Count:5
Secant Modulus 	0.901953 - 10.9027 GPa @Temperature -40.0 - 100 °C	130.819 - 1581.33 ksi @Temperature -40.0 - 212 °F	Average value: 5.67 GPa Grade Count:1
	0.901953 - 10.9027 GPa @Strain 0.190 - 10.2 %	130.819 - 1581.33 ksi @Strain 0.190 - 10.2 %	Average value: 5.67 GPa Grade Count:1
Izod Impact	29.0 - 603 J	21.4 - 445 ft-lb	Average value: 134 J Grade Count:20
Izod Impact, Notched	0.366 - 6.67 J/cm	0.685 - 12.5 ft-lb/in	Average value: 1.16 J/cm Grade Count:70
Izod Impact, Unnotched	3.67 - 5340 J/cm	6.88 - 10000 ft-lb/in	Average value: 8.93 J/cm Grade Count:53
Izod Impact, Notched (ISO)	1.00 - 88.2 kJ/m²	0.476 - 42.0 ft-lb/in²	Average value: 13.4 kJ/m² Grade Count:41

	4.40 - 20.0 kJ/m ² @Temperature -40.0 - -30.0 °C	2.09 - 9.52 ft-lb/in ² @Temperature -40.0 - -22.0 °F	Average value: 9.13 kJ/m ² Grade Count:8
Izod Impact, Unnotched (ISO)	4.00 - 21000 kJ/m ²	1.90 - 9990 ft-lb/in ²	Average value: 61.5 kJ/m ² Grade Count:32
	89.0 - 92.0 kJ/m ² @Temperature -30.0 - -30.0 °C	42.4 - 43.8 ft-lb/in ² @Temperature -22.0 - -22.0 °F	Average value: 90.3 kJ/m ² Grade Count:3
Charpy Impact Unnotched	3.00 - 2100 J/cm ²	14.3 - 9990 ft-lb/in ²	Average value: 8.73 J/cm ² Grade Count:137
	4.50 J/cm ² - NB @Temperature -40.0 - -20.0 °C	21.4 ft-lb/in ² - NB @Temperature -40.0 - -4.00 °F	Average value: 9.03 J/cm ² Grade Count:94
Charpy Impact, Notched	0.300 - 2100 J/cm ²	1.43 - 9990 ft-lb/in ²	Average value: 1.89 J/cm ² Grade Count:132
	0.300 - 11.0 J/cm ² @Temperature -40.0 - -20.0 °C	1.43 - 52.4 ft-lb/in ² @Temperature -40.0 - -4.00 °F	Average value: 1.53 J/cm ² Grade Count:102
Gardner Impact	4.63 - 36.2 J	3.42 - 26.7 ft-lb	Average value: 25.6 J Grade Count:3
Dart Drop, Total Energy	3.00 - 11.0 J	2.21 - 8.11 ft-lb	Average value: 6.92 J Grade Count:13
Falling Dart Impact	8.50 - 11.8 J	6.27 - 8.70 ft-lb	Average value: 9.88 J Grade Count:4
Instrumented Impact Total Energy	2.72 - 4.76 J	2.01 - 3.51 ft-lb	Average value: 3.55 J Grade Count:4
Coefficient of Friction	0.260 - 0.710	0.260 - 0.710	Average value: 0.484 Grade Count:9
Coefficient of Friction, Static	0.100 - 0.520	0.100 - 0.520	Average value: 0.373 Grade Count:9
K (wear) Factor	8.06 - 894 x 10 ⁻⁸ mm ³ /N-M	4.00 - 444 x 10 ⁻¹⁰ in ³ -min/ft-lb-hr	Average value: 125 x 10 ⁻⁸ mm ³ /N-M Grade Count:9
Taber Abrasion, mg/1000 Cycles	4.00 - 6.00	4.00 - 6.00	Average value: 5.67 Grade Count:6

Electrical Properties	Metric	English	Comments
Electrical Resistivity	5.00 - 1.00e+16 ohm-cm	5.00 - 1.00e+16 ohm-cm	Average value: 3.55e+14 ohm-cm Grade Count:93
Surface Resistance	10.0 - 1.00e+15 ohm	10.0 - 1.00e+15 ohm	Average value: 2.28e+13 ohm Grade Count:47
Dielectric Constant	3.10 - 3.90	3.10 - 3.90	Average value: 3.61 Grade Count:7
Dielectric Strength	5.00 - 45.0 kV/mm	127 - 1140 kV/in	Average value: 35.0 kV/mm Grade Count:57
Dissipation Factor	0.0140 - 0.0400	0.0140 - 0.0400	Average value: 0.0270 Grade Count:5
Comparative Tracking Index	450 - 600 V	450 - 600 V	Average value: 593 V Grade Count:41

Thermal Properties	Metric	English	Comments
CTE, linear	7.40 - 200 µm/m-°C	4.11 - 111 µin/in-°F	Average value: 68.7 µm/m-°C Grade Count:74

	18.0 - 100 µm/m-°C @Temperature 35.0 - 55.0 °C	10.0 - 55.6 µin/in-°F @Temperature 95.0 - 131 °F	Average value: 59.0 µm/m-°C Grade Count:2
	18.0 - 100 µm/m-°C @Thickness 2.00 - 2.00 mm	10.0 - 55.6 µin/in-°F @Thickness 0.0787 - 0.0787 in	Average value: 59.0 µm/m-°C Grade Count:2
CTE, linear, Transverse to Flow	48.0 - 170 µm/m-°C	26.7 - 94.4 µin/in-°F	Average value: 102 µm/m-°C Grade Count:60
Specific Heat Capacity	1.90 - 2.28 J/g-°C	0.454 - 0.545 BTU/lb-°F	Average value: 2.13 J/g-°C Grade Count:5
Thermal Conductivity	0.250 - 0.562 W/m-K	1.74 - 3.90 BTU-in/hr-ft²-°F	Average value: 0.359 W/m-K Grade Count:8
Melting Point	156 - 262 °C	313 - 504 °F	Average value: 221 °C Grade Count:153
Maximum Service Temperature, Air	70.0 - 220 °C	158 - 428 °F	Average value: 138 °C Grade Count:35
Deflection Temperature at 0.46 MPa (66 psi)	50.0 - 232 °C	122 - 450 °F	Average value: 181 °C Grade Count:102
Deflection Temperature at 1.8 MPa (264 psi)	40.0 - 227 °C	104 - 440 °F	Average value: 154 °C Grade Count:170
Deflection Temperature at 8.0 MPa	75.0 - 190 °C	167 - 374 °F	Average value: 135 °C Grade Count:25
Vicat Softening Point	180 - 221 °C	356 - 430 °F	Average value: 211 °C Grade Count:17
Glass Transition Temp, Tg	42.0 - 73.0 °C	108 - 163 °F	Average value: 55.2 °C Grade Count:16
UL RTI, Electrical	65.0 °C	149 °F	Average value: 65.0 °C Grade Count:7
UL RTI, Mechanical with Impact	65.0 °C	149 °F	Average value: 65.0 °C Grade Count:7
UL RTI, Mechanical without Impact	65.0 °C	149 °F	Average value: 65.0 °C Grade Count:7
Flammability, UL94	HB - V-0	HB - V-0	Grade Count:102
Flame Spread	80.0 mm/min	3.15 in/min	Average value: 80.0 mm/min Grade Count:9
Oxygen Index	24.0 - 25.0 %	24.0 - 25.0 %	Average value: 24.7 % Grade Count:3
Optical Properties	Metric	English	Comments
Yellow Index	8.00 %	8.00 %	Average value: 8.00 % Grade Count:3
Processing Properties	Metric	English	Comments
Processing Temperature	215 - 295 °C	419 - 563 °F	Average value: 279 °C Grade Count:20

Nozzle Temperature	240 - 310 °C	464 - 590 °F	Average value: 275 °C Grade Count:12
Die Temperature	210 - 270 °C	410 - 518 °F	Average value: 235 °C Grade Count:6
Melt Temperature	50.0 - 310 °C	122 - 590 °F	Average value: 267 °C Grade Count:124
Mold Temperature	8.00 - 120 °C	46.4 - 248 °F	Average value: 84.3 °C Grade Count:122
Drying Temperature	70.0 - 120 °C	158 - 248 °F	Average value: 81.3 °C Grade Count:96
Moisture Content	0.0200 - 0.250 %	0.0200 - 0.250 %	Average value: 0.152 % Grade Count:51
Dew Point	-17.8 °C	0.000 °F	Average value: -17.8 °C Grade Count:14
Injection Pressure	68.9 - 130 MPa	10000 - 18900 psi	Average value: 96.6 MPa Grade Count:31

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.